

Unit 04: Network Services and File Sharing

4.1 DNS and DHCP Configuration, Hostname Resolution, DNS Queries

1. DNS (Domain Name System)

DNS (Domain Name System) is a hierarchical and distributed naming system that translates **human-readable domain names** (like `www.example.com`) into **IP addresses** (like `192.0.2.1` or `2001:db8::1`) so computers can communicate over a network.

DNS was standardized by the **Internet Engineering Task Force (IETF)** in RFC 1034 and RFC 1035. It is one of the core components of the modern Internet.

Example

`www.google.com` → `142.250.x.x`

DNS Server Configuration (Linux – BIND)

Install BIND

```
sudo apt install bind9
```

Main Configuration File

```
/etc/bind/named.conf.local
```

Example Zone Configuration

```
zone "example.com" {  
    type master;  
    file "/etc/bind/db.example.com";  
};
```

Restart DNS Service

```
sudo systemctl restart bind9
```

DNS Queries & Testing

- Check DNS resolution: `nslookup google.com`

- Detailed DNS query: `dig google.com`
- Reverse lookup: `dig -x 8.8.8.8`

Hostname Resolution

Hostname resolution happens in this order:

1. `/etc/hosts` file
2. DNS server

View hostname

`hostname`

Change hostname

`sudo hostnamectl set-hostname server1`

Local resolution file

`/etc/hosts`

Example:

`192.168.1.10 server1`

2. DHCP (Dynamic Host Configuration Protocol)

DHCP automatically assigns:

- IP addresses
- Subnet masks
- Default gateway
- DNS servers

DHCP Server Configuration (Linux)

Install DHCP Server

`sudo apt install isc-dhcp-server`

Configuration File

/etc/dhcp/dhcpd.conf

Example Configuration

```
subnet 192.168.1.0 netmask 255.255.255.0 {  
    range 192.168.1.100 192.168.1.200;  
    option routers 192.168.1.1;  
    option domain-name-servers 8.8.8.8;  
}
```

Start Service

```
sudo systemctl start isc-dhcp-server
```

4.2 FTP, NFS, Samba Server Configuration, Directory Access

1. FTP (File Transfer Protocol)

FTP (File Transfer Protocol) is a **standard network protocol** used to transfer files between a **client** and a **server** over a TCP/IP network such as the Internet or a local network. It operates on a **client–server model**, where the client initiates a connection to upload, download, rename, or delete files on the remote server.

FTP was first defined in **Internet Engineering Task Force (IETF)** RFC 959 and works over **TCP**, typically using:

- **Port 21** → Control connection (commands and responses)
- **Port 20** → Data transfer (in active mode)

Install FTP Server (vsftpd)

```
sudo apt install vsftpd
```

Main Config File

/etc/vsftpd.conf

Start Service

```
sudo systemctl start vsftpd
```

Secure FTP Best Practices

- Disable anonymous login
- Use FTPS (SSL/TLS encryption)
- Restrict users to home directories

2. NFS (Network File System)

NFS (Network File System) is a distributed file system protocol that allows a computer to **access files and directories over a network as if they were stored locally**. It is mainly used in **Linux/Unix environments** to share directories between servers and clients.

NFS was originally developed by **Sun Microsystems** in 1984 and is now maintained as an open standard by the **Internet Engineering Task Force (IETF)**.

Install NFS Server

```
sudo apt install nfs-kernel-server
```

Configure Shared Directory

Edit:

```
/etc/exports
```

Example:

```
/shared 192.168.1.0/24(rw,sync,no_subtree_check)
```

Apply Changes

```
sudo exportfs -a
```

3. Samba Server (SMB Protocol)

Samba is an open-source software suite that enables **file and printer sharing between Linux/Unix systems and Windows systems** using the **SMB (Server Message Block) protocol**, also known as **CIFS (Common Internet File System)**.

Samba implements the SMB protocol originally developed by **IBM** and later extended by **Microsoft** for Windows networking. The Samba project itself is maintained by the **Samba Team**.

Install Samba

```
sudo apt install samba
```

Configuration File

```
/etc/samba/smb.conf
```

Example Share

```
[shared]  
path = /shared  
browseable = yes  
writable = yes  
guest ok = no
```

Restart Service

```
sudo systemctl restart smbd
```

Best Practices for Secure File Sharing

- ✓ Use strong authentication
- ✓ Disable anonymous access
- ✓ Apply proper directory permissions (chmod, chown)
- ✓ Use firewall rules to restrict access
- ✓ Encrypt data transmission (FTPS, SMB3 encryption)
- ✓ Regularly update services

Directory Access Control

Set Permissions

```
chmod 755 directory  
chown user:group directory
```

Use ACL for Fine-Grained Control

```
setfacl -m u:username:rwX directory  
getfacl directory
```